

Pragatheeswaran K

Senior Software Engineer | React.js | Next.js | TypeScript
(6+ years exp)

Senior Front-End Engineer with 6+ years of experience delivering performant, production-grade applications using React, Next.js, and TypeScript. Specialized in front-end architecture, rendering pipelines, and browser internals, with a strong focus on performance, accessibility, and scalability. Proven experience building and maintaining complex UI systems in fintech, healthcare, and ed-tech domains. Trusted to drive features end-to-end — from technical design to final delivery.

Contacts

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<https://pragatheeswaran.vercel.app/>

SKILLS

Frontend: HTML/CSS, Javascript, Typescript, ReactJS, Nodejs.

Frameworks & Libraries:
React.js, Next.js, Material-UI (MUI), Tailwind CSS, Redux, Zustand, Lodash

Testing: Jest, React Testing Library (RTL)

Architectural & Design:
Front-End System Design, Component-Based Architecture, Performance Optimization (Web Vitals, Lighthouse), Scalability, Accessibility (A11y)

Version Control: Git (GitHub, GitLab)

Tools: Webpack, Vite

EXPERIENCE

Significant Highlights

- 6+ years of professional experience building high-performance web applications using React.js, Next.js, TypeScript, Node.js, Material UI, and Tailwind CSS.
- Contributed to the JavaScript ecosystem by publishing a utility-focused NPM package (**pragi-string**) that handles complex string and number formatting.
- Led key front-end initiatives in the **MSCIQ (Financial Planning & Analysis)** project, improving data handling and spreadsheet performance using Jspreadsheet and custom UI components.
- Delivered core modules for Terv, including AI-powered proctoring using TensorFlow.js to detect and capture student malpractice during assessments, along with interactive tools that significantly improved platform integrity and user experience.
- Took ownership of **SPICE 2.0** frontend redevelopment—migrated to modern React, removed deprecated dependencies, and introduced a scalable architecture with dynamic form support.

Professional Experience

Ideas2IT (Mar 2023 – Present)

Projects: SPICE, MSCIQ

Role: Senior Software Engineer → Technical Analyst

◆ SPICE – Digital Healthcare Platform

Domain: Healthcare | Open Source | Public Health
SPICE is an open-source digital healthcare platform focused on clinical and administrative workflows in low-resource settings.

Community Contribution

NPM Package :

<https://www.npmjs.com/package/pragi-string>

Medium Blogs:

1. [Deep Dive into React’s Diffing and Reconciliation Process](#)

2. [Time and space complexity](#)

3. [DSA for frontend developers](#)

EDUCATION

B.E in Computer Science Engineering, from Dhanalakshmi Srinivasan of Engineering and Technology, Chennai.

- Contributed to SPICE 1.0 and later led front-end development for SPICE 2.0 with a complete overhaul.
- Re-architected and rebuilt the entire front-end from scratch with a modern, scalable architecture.
- Refactored legacy code into reusable modules and adopted scalable component-based architecture.
- Audited and replaced deprecated packages for long-term stability.
- Upgraded React to the latest stable version, aligning with best practices.
- Integrated a dynamic form engine rendering reusable UI based on FHIR-mapped JSON.

Tech Stack: ReactJS, TypeScript, Redux-Saga, React Testing Library (RTL), FHIR, Material-UI

◆ MSCIQ – Financial Planning & Analysis Tool

Domain: FinTech / Enterprise Data Tools

MSCIQ is a strategic financial planning tool that enables finance teams to streamline data modeling and decision-making.

- Spearheaded spreadsheet-like UI for finance use cases using React + Js spreadsheet.
- Integrated complex, nested datasets with MUI TreeView for intuitive navigation.
- Implemented modular architecture and state management using Zustand.
- Delivered a responsive UI with highly reusable components tailored for enterprise needs.

Tech Stack: ReactJS, TypeScript, Zustand, Jest, MUI, Js spreadsheet, Lodash

◆ LegacyLeap – AI IDE Extension for Legacy Code Modernization

Domain: Developer Tools | AI-Enhanced IDE

LegacyLeap is an AI-driven extension to help developers modernize legacy codebases with intelligent suggestions.

- Developed a real-time Code Diff View comparing AI-suggested transformations with the original code.
- Created an “Impact View” heatmap to visualize legacy hotspots.

- Integrated deeply into VSCode via Extension API with optimized runtime performance.

Tech Stack: TypeScript, VSCode Extension API, AI Model Integration, CSS Modules

Techfully (Aug 2021 – Mar 2023)

Role: Front-End Developer

Projects: Terv, DocsAuth

◆ Terv – Online Assessment & Learning Platform

Domain: EdTech

Terv is a platform that enables students to participate in live assessments and interactively consume educational content.

- Implemented **AI-powered proctoring** using TensorFlow.js to detect anomalies and ensure assessment integrity.
- Developed a **custom video player** with progress tracking and an integrated PDF viewer for study materials.
- Delivered **accessible, responsive UI** while collaborating cross-functionally with backend and product teams.

Tech Stack: ReactJS, Redux Toolkit, TypeScript, Material-UI (MUI), TensorFlow.js

◆ DocsAuth – Certificate Application Platform

Domain: EdTech

DocsAuth simplifies the online certificate application process with a dynamic and user-friendly interface.

- Implemented a **JSON-driven dynamic form engine** that renders various input types based on configuration, allowing clients to build custom forms without code changes.
- Focused on modular, maintainable front-end architecture to scale across various institutions.

Tech Stack: ReactJS, Redux Toolkit, TypeScript, Material-UI (MUI), Node.js.

Padink Engineering Service (Jun 2019 – Jul '21)

Role: Front-End Developer

Project: Padink Engineering Tracker

◆ Padink Engineering Tracker

Domain: Engineering / Industrial Design

Padink Engineering Tracker is an internal tool designed to track CAD and SP3D design workflows along with employee task and performance management.

- Built interfaces to **monitor project progress** across multiple design systems like CAD and SP3D.
- Developed modules for **employee management**, including assignment tracking, daily logs, and reporting dashboards.
- Focused on **streamlined UI/UX** for non-tech users in the engineering domain.

Tech Stack: ReactJS, JavaScript, Bootstrap, REST APIs